

From: Braum, Thomas (EGLE)
Sent: 9/13/2023 1:21:16 PM
To: "Barbieri, Charles" <CBarbieri@fosterswift.com>
Cc: "Staron, Matthew (EGLE)" <STARONM@michigan.gov>; "Brady, Donal (EGLE)" <BradyD6@michigan.gov>; "Schmitt, Glen (EGLE)" <SCHMITTG1@michigan.gov>; "Lowe, Claire (EGLE)" <LoweC9@michigan.gov>; "Cindy Tomaszewski" <ctomaszewski@Cadillac-MI.net>; "Jeff Dietlin" <utilities@cadillac-mi.net>; "Quinn, Patrick" <pquinn@fosterswift0.onmicrosoft.com>; "DePetro, Phillip (EGLE)" <DePetroP1@michigan.gov>
Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit
Attachments: Ammonia implemetation memo final updated 2023.pdf

Hi Chuck,

Thank you for your response. Please let me know when the City has had adequate time to review the draft permit further and determined whether any items require additional discussion.

As previously communicated, the ammonia nitrogen certification requirement has been included as an opportunity for the facility to evaluate whether revised ammonia nitrogen limitations based on new chronic toxicity criterion that will be imposed can be met before the limits become effective. The Department has developed this strategy to allow facility operational changes or permit modifications to be completed if needed, rather than requiring these limits be achieved upon the effective date of the reissued permit. This evaluation may or may not require outside review; it will be up to each facility to determine how this certification will be completed.

If the evaluation supports negative certification, the permit may be modified to include a schedule of compliance developed by the Department for the facility to meet the revised limits in accordance with the attached memo outlining the implementation of the new ammonia criterion. The ammonia nitrogen limits would take effect at the completion of this schedule. The development of the schedule will consider the useful life of the wastewater treatment plant and an updated Financial Capability Analysis (FCA) submitted by the City. If the City adheres to the compliance dates included in the permit regarding ammonia nitrogen certification, noncompliance actions should not occur.

Thank you for providing comments regarding this draft permit and please let us know if you have any additional questions, comments, or concerns.

Sincerely,

Tom Braum

Environmental Quality Analyst

Municipal Permits Unit

Water Resources Division

Michigan Department of Environment, Great Lakes, and Energy

BraumT2@michigan.gov | 517-331-7377

From: Barbieri, Charles <CBarbieri@fosterswift.com>
Sent: Tuesday, September 12, 2023 2:09 PM
To: Braum, Thomas (EGLE) <BraumT2@michigan.gov>

Cc: Staron, Matthew (EGLE) <STARONM@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Lowe, Claire (EGLE) <LoweC9@michigan.gov>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>; Jeff Dietlin <utilities@cadillac-mi.net>; Quinn, Patrick <pquinn@fosterswift0.onmicrosoft.com>; DePetro, Phillip (EGLE) <DePetroP1@michigan.gov>

Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

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9-12-23

Tom, we conferred today internally, and we acknowledge and appreciate the response and proposed revisions. Although the City staff wanted to have a chance to review the draft permit further, they have asked me in the meantime to raise their concern again about the ammonia certification, as that type of certification will likely require outside review. Beyond that, the City would like to know what would be the Department's best prediction on how it would react if the certification to meet the ammonia nitrogen limits could not be met. The suggestion is made on page 8 of the permit that the Department may be modified. Can you advise on that point about what that means? And would there be any other action (e.g. an issue of noncompliance, etc.) contemplated by the Department at that point? The City wants a clear understanding what a negative certification could mean? After receiving your additional thoughts, we can complete the review. Thanks. Chuck

Charles E. Barbieri

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FOSTER SWIFT
FOSTER SWIFT COLLINS & SMITH PC ■ ATTORNEYS

From: Braum, Thomas (EGLE) <BraumT2@michigan.gov>

Sent: Friday, September 8, 2023 10:08 AM

To: Barbieri, Charles <CBarbieri@fosterswift.com>

Cc: Staron, Matthew (EGLE) <STARONM@michigan.gov>; Brady, Donal (EGLE)

<BradyD6@michigan.gov>; Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Lowe, Claire (EGLE)

<LoweC9@michigan.gov>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>; Jeff Dietlin

<utilities@cadillac-mi.net>; Quinn, Patrick <pquinn@fosterswift0.onmicrosoft.com>; DePetro, Phillip

(EGLE) <DePetroP1@michigan.gov>

Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

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Chuck,

Based on our discussion held on August 29th, we have elected to make the following revisions to the draft NPDES permit for the Cadillac WWTP. The attached draft documents reflect these revisions.

1. The monitoring frequencies for Total Arsenic, Total Beryllium, Total Cadmium, Total Silver, Total Copper, and Total Benzidine have been revised from weekly to twice monthly.
2. The timing of the option to submit a monitoring frequency reduction request for Available Cyanide, Total Arsenic, Total Benzidine, Total Beryllium, Total Bromomethane, Total Cadmium, and/or Total Silver has been reduced from 24 months to 12 months. The monitoring frequency for Total Arsenic, Total Beryllium, Total Cadmium, and Total Silver shall not be reduced to less than quarterly.
3. The daily maximum loading reporting requirement for Total Copper and Total Bromomethane has been removed.
4. The Acute Toxicity limit of 1.0 TU_A has been revised to a reporting requirement. Part I.A.1.j. has been revised in accordance with this change.
5. A requirement to conduct a survey of the facility's Significant Industrial Users has been added to Part I.C.1. Michigan Industrial Pretreatment Program. This requirement has been added as part of the decision to reduce the monitoring frequency and timeframe for a monitoring frequency reduction request for the toxic parameters listed above to ensure the sources of these pollutants are identified and efforts to eliminate them are made.

Per our discussion, the Conventional Water Quality-Based Effluent Limit (WQBEL) Review is also attached. This review provides final effluent recommendations for CBOD5, Ammonia Nitrogen, and Dissolved Oxygen. The conventional pollutant items below were addressed during our call.

- The final effluent limitations for CBOD5 and Ammonia Nitrogen based on permit writer judgement during the October through November season.
- The revised monthly average effluent limitation for CBOD5 for the December through March season.
- The revised loading limitations for CBOD5 and Ammonia Nitrogen.

The City requested further clarification regarding our calculation process for loading limitations and the translation of WQBEL recommendations to monthly average limits for CBOD5.

Loading limits are determined using a calculation based on the facility's design flow, the parameter's concentration limit, and the weight of a gallon of water. The formula to determine a

loading limit is: Design Flow * Concentration Limit * 8.34 = Loading Limit. The 7-day average loading limit for CBOD5 during the October through November season was revised from 480 lbs/day to 450 lbs/day. This value was determined using the formula above, 3.2 MGD * 17mg/l * 8.34 lbs = 453.696 lbs/day, which is then rounded to two significant figures resulting in the 450 lbs/day limit. The same process was used to revise the monthly average loading limit for Ammonia Nitrogen during the October through November season from 88 lbs/day to 43 lbs/day (3.2 MGD * 1.6 mg/l * 8.34 = 42.7 lbs/day, rounded to 43 lbs/day). The previously included loading limitations for these seasons were determined to be errors as this is our standard procedure for calculating these limits for all NPDES permits.

The 34 mg/l daily maximum CBOD5 limit for the December through March season was a previously recommended WQBEL. It is the permit writer's responsibility to translate daily maximum CBOD5 limit recommendations to monthly average limitations. Daily maximum CBOD5 limit recommendations have correlating monthly average limitations that are incorporated into NPDES permits. The correlating monthly average limit for a daily maximum limit recommendation of 34 mg/l for CBOD5 is 23 mg/l. This value is consistent with the procedure used for CBOD5 limits statewide. It was determined that the previously included monthly average limit of 25 mg/l needed to be revised to remain consistent with our permitting procedure.

Lastly, the City requested information regarding our determination on the facility's ability to meet the revised ammonia nitrogen limitations developed using the new chronic toxicity criterion for Ammonia Nitrogen adopted by the Department in 2019. When developing WQBEL recommendations for Ammonia Nitrogen using the new criterion, facilities were grouped into one of three categories: facilities not receiving more stringent limits, facilities receiving more stringent limits that can be met based on existing effluent quality, and facilities receiving more stringent limits that cannot currently be met. This determination was made using existing effluent quality and a 95th percentile calculation. We consider a monthly average limit achievable if the 95th percentile of the existing effluent data is less than the WQBEL.

For Cadillac, the ammonia nitrogen WQBEL was more stringent than the effluent limit in the facility's current permit. Effluent data from a five-year period spanning from January 2017 through January 2022 was used to determine if the facility could meet the more stringent limits. Looking at the 95th percentile of Cadillac's effluent data, the facility is capable of meeting the revised effluent limits at their current flow rate. The calculations used to make this determination are attached. For facilities receiving more stringent effluent limits that can currently be met, we are including the ammonia nitrogen certification condition to allow the facility to determine whether the limits would also be able to be met if flows increase to the facility's design flow.

Please let us know if you have any additional questions, comments, or concerns based on these responses and the revised documents. Thank you and I look forward to hearing back from you.

Sincerely,

Tom Braum

Environmental Quality Analyst

Municipal Permits Unit

Water Resources Division

Michigan Department of Environment, Great Lakes, and Energy

BraumT2@michigan.gov | 517-331-7377

From: Staron, Matthew (EGLE) <STARONM@michigan.gov>

Sent: Tuesday, August 29, 2023 4:33 PM

To: Barbieri, Charles <cbarbieri@fosterswift.com>; Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Lowe, Claire (EGLE) <LoweC9@michigan.gov>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>; Jeff Dietlin <utilities@cadillac-mi.net>; Quinn, Patrick <pquinn@fosterswift0.onmicrosoft.com>

Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

Thanks Chuck. I agree that it was a good dialogue. We should have a response later this week or early next.

Matt.

From: Barbieri, Charles <CBarbieri@fosterswift.com>

Sent: Tuesday, August 29, 2023 4:22 PM

To: Staron, Matthew (EGLE) <STARONM@michigan.gov>; Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Lowe, Claire (EGLE) <LoweC9@michigan.gov>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>; Jeff Dietlin <utilities@cadillac-mi.net>; Quinn, Patrick <pquinn@fosterswift0.onmicrosoft.com>

Subject: FW: Preliminary City of Cadillac Comments on New Draft NPDES permit

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8-29-23

Folks, thanks again for the dialogue this afternoon. While the matter is still fresh, Cindy Tomaszewski of Cadillac has forwarded the email chain that she had with the laboratory on the anomalous benzidine result and follow-up analyses that occurred. See first attachment with highlighting. She also provided the follow-up lab analyses for the additional sampling collected and split on 8-31-21. See second and third attachments. This should validate the additional follow-up that occurred and the City's concern about the anomalous result of benzidine. We look forward to your additional evaluation of the items discussed. Advise if you have any questions. Chuck

Charles E. Barbieri

Attorney

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313 South Washington Square

Lansing, MI 48933-2193

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cbarbieri@fosterswift.com
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From: Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Sent: Tuesday, August 29, 2023 4:12 PM
To: Jeff Dietlin <utilities@cadillac-mi.net>; Barbieri, Charles <CBarbieri@fosterswift.com>; Patrick Miller <pmiller@cadillac-mi.net>
Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

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Email string and sample data from 8/31/21

Cindy A. Tomaszewski
Lab Supervisor

From: Jeff Dietlin <utilities@cadillac-mi.net>
Sent: Tuesday, August 29, 2023 3:29 PM
To: Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>; 'Barbieri, Charles' <CBarbieri@fosterswift.com>; Patrick Miller <pmiller@cadillac-mi.net>
Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

I can't open this.

Jeffrey M. Dietlin
City of Cadillac
Director of Utilities
(231) 779-7346(Direct)
(231)775-0906 (Fax)

From: Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Sent: Tuesday, August 29, 2023 3:25 PM
To: 'Barbieri, Charles' <CBarbieri@fosterswift.com>; Jeff Dietlin <utilities@cadillac-mi.net>; Patrick Miller <pmiller@cadillac-mi.net>
Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit
Importance: High

FYI regarding benzidine. You have to drill through a few layers to get to the email string.

Cindy A. Tomaszewski
Lab Supervisor

From: Barbieri, Charles <CBarbieri@fosterswift.com>

Sent: Friday, August 25, 2023 2:03 PM

To: Jeff Dietlin <utilities@cadillac-mi.net>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>; Patrick Miller <pmiller@cadillac-mi.net>

Subject: Fwd: Preliminary City of Cadillac Comments on New Draft NPDES permit

Folks, what date or dates do you prefer for follow-up call? Chuck

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From: Braum, Thomas (EGLE) <BraumT2@michigan.gov>

Sent: Friday, August 25, 2023 11:06:53 AM

To: Barbieri, Charles <CBarbieri@fosterswift.com>

Cc: Brady, Donal (EGLE) <BradyD6@michigan.gov>; Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>; Jeff Dietlin <utilities@cadillac-mi.net>; Patrick Miller <pmiller@cadillac-mi.net>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>; Staron, Matthew (EGLE) <STARONM@michigan.gov>; Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Lowe, Claire (EGLE) <LoweC9@michigan.gov>

Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

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Good morning Chuck,

Our responses to the preliminary comments provided from the City regarding the Cadillac WWTP NPDES Permit No. MI0020257 are attached. Also attached is a revised draft of the NPDES permit based on our response to the Industrial Pretreatment Program item provided.

We would like to set up a time to meet next week to discuss additional questions or comments the City may have. Please let me know if any of the following times would work for a meeting:

- Tuesday August 29th at 9:00 AM
- Tuesday August 29th at 2:00 PM
- Tuesday September 5th at 8:30 AM
- Friday September 8th at 11:00 AM

Thank you and we look forward to hearing back from you.

Sincerely,

Tom Braum

Environmental Quality Analyst

Municipal Permits Unit

Water Resources Division

Michigan Department of Environment, Great Lakes, and Energy

BraumT2@michigan.gov | 517-331-7377

From: Barbieri, Charles <CBarbieri@fosterswift.com>
Sent: Tuesday, August 22, 2023 11:37 AM
To: Staron, Matthew (EGLE) <STARONM@michigan.gov>
Cc: Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>; Jeff Dietlin <utilities@cadillac-mi.net>; Patrick Miller <pmiller@cadillac-mi.net>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

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Thank you for the update.

Charles E. Barbieri

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From: Staron, Matthew (EGLE) <STARONM@michigan.gov>
Sent: Tuesday, August 22, 2023 11:32 AM
To: Barbieri, Charles <CBarbieri@fosterswift.com>
Cc: Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>; Jeff Dietlin <utilities@cadillac-mi.net>; Patrick Miller <pmiller@cadillac-mi.net>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

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Hi Chuck.

We are preparing responses to your written comments. Once we send those out and give you and your client a chance to review them, we can set up a time to meet. I appreciate your understanding in our goal to issue the permit by Oct. 1.

Matt.

From: Barbieri, Charles <CBarbieri@fosterswift.com>
Sent: Tuesday, August 22, 2023 11:24 AM
To: Staron, Matthew (EGLE) <STARONM@michigan.gov>
Cc: Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>; Jeff Dietlin <utilities@cadillac-mi.net>; Patrick Miller <pmiller@cadillac-mi.net>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Subject: RE: Preliminary City of Cadillac Comments on New Draft NPDES permit

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8-22-23

Matt, I wanted to see if we can set up a time this week to have the follow-up call on this, given the schedule the Department has set. Do you have some possible dates and times we can shoot for? Thanks. Chuck

Charles E. Barbieri

Attorney

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From: Barbieri, Charles
Sent: Friday, August 18, 2023 3:31 PM
To: Staron, Matthew (EGLE) <STARONM@michigan.gov>
Cc: Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>; Jeff Dietlin <utilities@cadillac-mi.net>; Patrick Miller <pmiller@cadillac-mi.net>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Subject: Preliminary City of Cadillac Comments on New Draft NPDES permit

8-18-23

Matt, at the risk of being unable to understand some of the comments that were discussed internally and with the expectation that we can have a follow-up remote meeting that includes my client representatives, let me offer this preliminary description of issues that the City has about some of the permit items. We would like to see if some of these concerns could be eliminated now so that

additional comments on the permit are not needed. Please let us know when we can discuss next week. Thanks. Chuck

Charles E. Barbieri

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Ammonia implemetation memo final updated 2023.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

VIA EMAIL

TO: Chris Alexander, Manager, Permits Section, Water Resources Division (WRD)
Mike Alexander, Manager, Surface Water Assessment Section, WRD
Matt Staron, Supervisor, Municipal Permit Unit, Permits Section, WRD
Susan Doty, Supervisor, Enforcement Unit,
Field Operations Support Section, WRD
Charlie Hill, Design Engineers Committee Chair, WRD
Chris Veldkamp, NPDES Committee Chair, WRD

FROM: Phil Argiroff, Assistant Director, Water Resources Division

DATE: April 27, 2023

SUBJECT: Factors for Water Resources Division Staff to Consider When
Implementing the 2019 Quality Values for Ammonia Update

Phil Argiroff

The purpose of this memo is to identify the factors Water Resources Division (WRD) staff should consider when developing schedules of compliance in National Pollutant Discharge Elimination System (NPDES) permits to implement the revised water quality values for ammonia that EGLE developed in 2019 pursuant to Rule 323.1057, Toxic Substances (Rule 57), of the [Part 4 Water Quality Standards](#), promulgated Pursuant to Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended (NREPA). Ammonia toxicity water quality values (WQVs) have been developed and revised periodically under Rule 57 since WRD was delegated the NPDES Program in 1973. The United States Environmental Protection Agency (USEPA), Region 5 has requested that EGLE revise the ammonia WQVs since 2013, and WRD is now implementing the revised WQVs in our surface water monitoring/assessment program and in National Pollutant Discharge Elimination System (NPDES) permits. The revised WQVs may result in more stringent water quality-based effluent limits to protect against chronic toxicity in all seasons, especially the colder seasons. The revised WQVs were finalized by WRD in 2019 after the federal Clean Water Act (CWA) was amended in January 2019 to add Section 402(s) for integrated planning, [\[33 USC 1342\(s\)\]](#). Section 402(s) allows implementation of a new water quality standard (or WQV) in NPDES permits over several permit cycles.

This memo describes the basic factors that should be considered to set schedules of compliance in NPDES permits, or in more limited situations, in administrative consent orders (ACOs). Schedules of compliance for water quality-based effluent limits for ammonia may be placed in NPDES permits for existing facilities when permits are reissued after October 1, 2022. For existing municipal wastewater treatment plants

SUBJECT: Factors for Water Resources Division Staff to Consider when Implementing the 2019 Quality Values for Ammonia Update

Page 2

April 27, 2023

(WWTP), staff is asked to consider all appropriate factors listed below and set final schedules of compliance based on case-by-case judgment of these factors.

For new municipal WWTPs and industrial facilities, the revised WQVs will be used when permits are issued (even with permits issued before October 1, 2022). This will ensure facilities are designed appropriately. For existing industrial facilities, schedules to meet final limits (as necessary) can be set without considering the items below, but instead on achieving maximum but reasonable progress. Staff are directed to consider Rule 323.2153, Point source discharges; standards of performance (Rule 53), of the [Part 21, Wastewater Discharge Permits \(Part 21 Rules\)](#), promulgated pursuant to Part 31, Water Resources Protection, of the NREPA, for additional guidance for existing facilities. Note that to evaluate water quality, the WRD's Surface Water Assessment Section can use the revised WQVs immediately. Staff should evaluate the following factors below when determining an appropriate schedule of compliance for municipal WWTPs:

Schedules of compliance should consider:

- For new and upgrading/expanding WWTPs, ammonia toxicity-based permit limits can be required at the start-up of the new or upgraded/expanded WWTP. This can start before WRD begins to implement the revised ammonia WQVs for existing WWTPs starting on October 1, 2022.
- For existing WWTPs, analyze sufficient effluent ammonia data and determine if the WWTP can comply with the revised limits. Allow a two-year evaluation by the permittee to verify that they can comply with the revised limits at annual average design flow, and consider other factor involving nitrification. If verified, the revised limit become effective after three years. If not able to meet or the permittee does not verify, then consider the life of a WWTP before requiring compliance with any revised ammonia toxicity-based permit limits. For depreciation purposes, the life of a WWTP is typically 20 years. Also consider Rule 53 of the [Part 21 Rules](#), promulgated pursuant to Part 31, Water Resources Protection, of the NREPA, which requires, at a minimum, a 10-year schedule from when a previous requirement was set for a pollutant. Further, consider when the last upgrade/expansion was made to aeration units at the WWTP as these process units are most directly involved with meeting ammonia limits.
- For existing WWTPs, ask for a completed Financial Capability Analysis ([Clean Water Act Financial Capability Assessment Guidance](#)). Based on the final determination of financial burden for the municipality (low, medium, or high), attempt to keep the permittee at the same classification of burden (low, medium, or high) if any modifications are needed to meet water quality-based effluent limits based on the revised WQVs. This will affect the length of schedule of

SUBJECT: Factors for Water Resources Division Staff to Consider when Implementing the 2019 Quality Values for Ammonia Update

Page 3

April 27, 2023

compliance and/or the sequencing of the schedule. As an outside boundary, consider setting a schedule of compliance for a maximum of 20 years if the result for the upgraded WWTP is low or medium. Consider setting a 25-year schedule if the final burden is high.

- For sequencing of federal CWA requirements, consider an integrated plan. Even if the plan does not include all requirements specified by the USEPA, use the concepts contained in the USEPA's Web site (epa.gov/npdes/integrated-planning-municipal-stormwater-and-wastewater), to set schedules of compliance in permits so that the federal CWA's requirements can be sequenced. WRD's priority is to set schedules that first address existing combined sewer overflow or sanitary sewer overflow control programs, Infiltration/Inflow reduction efforts, or elimination of bypassing or blending at the WWTP. Then address the need for upgraded treatment or operational control at the WWTP to meet the revised ammonia WQVs. In some cases, an expanded WWTP might be necessary to control wet weather flows. If this is the case, then it may make sense to require compliance with the revised WQV at the same time as an upgraded WWTP.
- There may also be an opportunity to use Utility of the Future (i.e., Water Resource Recovery Facility [WRRF]) concepts to help address additional nitrification needs at existing WWTPs. This might allow for schedules to proceed at a quicker pace. As an example, the WRRF concepts might use available carbon at the WWTP for energy generation, and the remaining WWTP aeration capacity might be sufficient to provide sufficient nitrification. District engineers should work with permittees to explore some possible treatment solutions that may allow schedules to proceed faster and/or at a lower capital and operational cost.

Implementation of the revised WQVs for ammonia is important to protect surface water quality. However, reasonable schedules to comply with the effluent limits to meet the WQVs are critically important given other responsibilities that permittees have under the federal CWA, and potentially limited dollars for infrastructure. It is WRD's desire that this memo will help staff set appropriate and cost-effective schedules to meet the revised ammonia WQVs.